

Preliminary Report

Project Management

ECLift is a student engineering team founded in late 2023 at the French university of École Centrale de Lyon, with the goal of participating in the second edition of the XtraChallenge. This diverse team is made up of thirteen international students of the Aerospace Engineering Master's and divided into five departments in order to better organize the execution of the different tasks.

The project development involves efficient management of human and material resources. For his purpose, a tight schedule and a good communication between departments has been key to the success of the design phase of the prototype: *ECLifter*.

Design Process

Propulsion

The propulsive requirements are probably the most influential factor of all. The propeller-motor array was studied both theoretically and experimentally by means of a static test. The available thrust was then used to sketch an aircraft with the right proportions, and aerodynamic capabilities accordingly.

Aerodynamics

The aerodynamic design process involved an effort to balance the aircraft's capabilities in order to achieve a good overall scoring performance. The main focus was lift optimization, aiming for a high payload capacity and possibly the bonus given for short take off, which resulted in the selection of the S1221 airfoil for the main wing. The inexperience of the pilot was also considered, which is seen reflected in certain choices for the general layout (Figure 2) and also during the stability analysis (Tables 1 and 2), which help with maneuverability and reactivity of the aircraft to the controller inputs. XFLR5 and Star CCM+ were used to obtain all the relevant data.

	$CM(a=0)$	CMa	CLb	CNb
Criteria	>0	<0	<0	>0
ECLifter	0.017	-2.2039	-0.022295	0.060894

Table 1: Static stability derivatives.

	$CM\delta e$	$CL\delta a$	$CN\delta a$	$CL\delta r$	$CN\delta r$
Criteria	<0	>0	≈ 0	>0	<0
ECLifter	-1.362	0.13828	0.000252	0.004345	-0.044582

Table 2: Dynamic stability derivatives.

Structures

The structural development began with a thorough read of the competition regulations leading to material selection and literature reviews of similar aircraft. The number of balls to be carried, enforced box length, and avionics size the fuselage's length and diameter. With this general length/diameter and the application DevFuse, the fuselage structure was modeled. Furthermore, all lifting surfaces are modeled after aerodynamic conclusions. The complete 3D model of the aircraft considers the load paths, avionics layout, cargo storage, connection points, and control surfaces. Most importantly, the aircraft is bound by the size requirements of the loading box. In the following section, the model and the general layout of the aircraft is presented.

Aircraft Configuration

The aircraft has a high wing with taper, divided into four panels for disassembly. A conventional tail is used for structural reliability. The main philosophy has been to simplify the design whenever possible. The landing gear, joiners, and tail wheel are off-the shelf. Each wing panel has a single movable surface that extends along its entire span. The flaps are thus contained within the inboard panels, and the ailerons within the outboard ones.

The *ECLifter* should have an OEW just below 2kg upon manufacturing and be able to take off with a minimum of 6 billiard balls and a maximum of 10, while accommodating at least 25 ping-pong balls. The locations of payload, avionics, and connection sites are shown in Figure 4. Avionics refers to a main bay (holds battery, receiver, ESC, and motor). The connection sites refer to areas that will see spar housing or screws/glues once final assembly commences.

Manufacturing

The manufacturing has begun with the wing assembly to be followed by the fuselage and empennage. The bulk of the wood cutting is done using a laser cutter for ease of process and accuracy. In Figure 5 an example is given; the ribs of the main wing are being cut. The ribs all have labeling carved into them to ease the assembly process. In Figure 6 the balsa leading edge is attached to the wing assembly. PVC tubes are added to allow for spar guidance. For now, wood glue is used for all surface connections, however it will be replaced by CA glue in many places on the final airplane. For the fuselage, plans and manufacturing lists have been prepared. They include the size of the part, its material, and any special considerations to be noted. The spar-body connection is being handled by 3-D printed housings. Lastly, in Figure 7 a perspective-view of the plane is shown to illustrate what the partial assembly of the *ECLifter* will look like upon the completion of manufacturing.

Challenges and Next Steps

As of today, all the focus is on completing a working prototype of the aircraft. With the wing nearing completion, the cutting and gluing of the fuselage is soon to follow. This will allow the team to then make notes of the drawbacks the aircraft possess and make plans for improvement on future iterations. Challenges going forward will include streamlining of manufacturing methods and development of best practices regarding wooden RC aircraft manufacturing.

Appendix: Figures

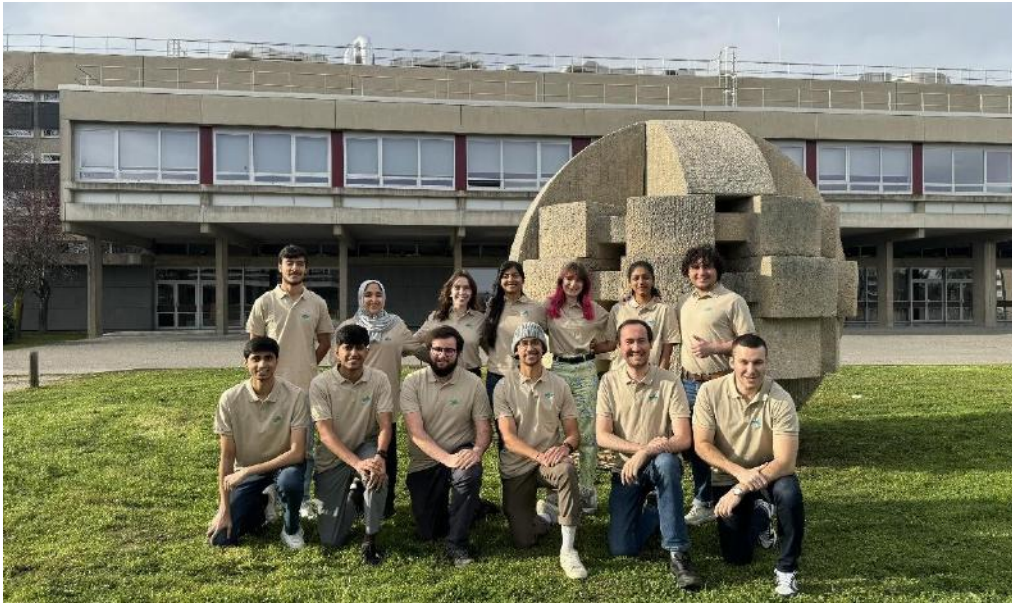


Figure 1: ECLifter gathers team members from seven nationalities and many engineering disciplines.

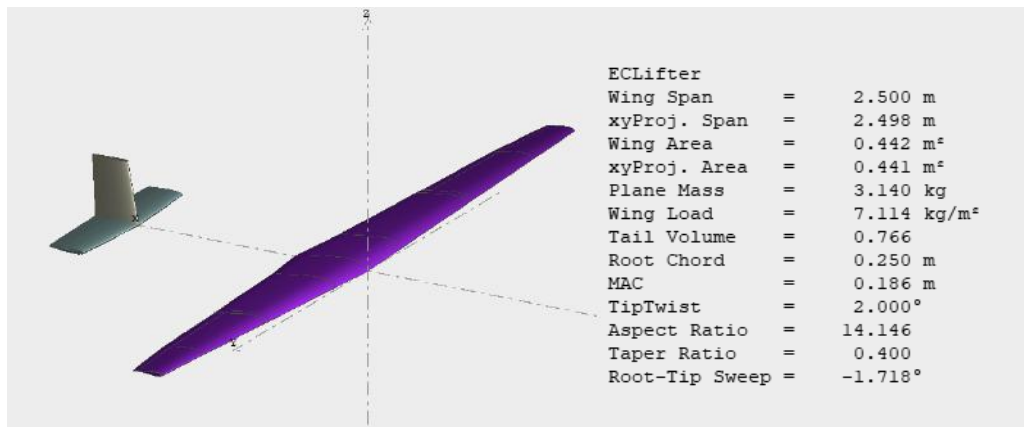


Figure 2: Wing and empennage geometry of the ECLifter in XFLR5 together with relevant parameters.

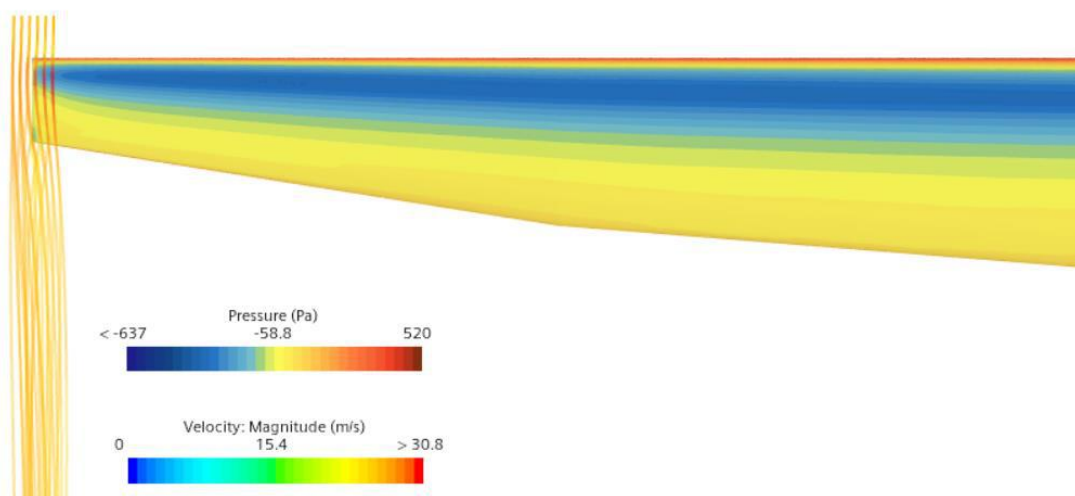


Figure 3: CFD preliminary study of the wing geometry.

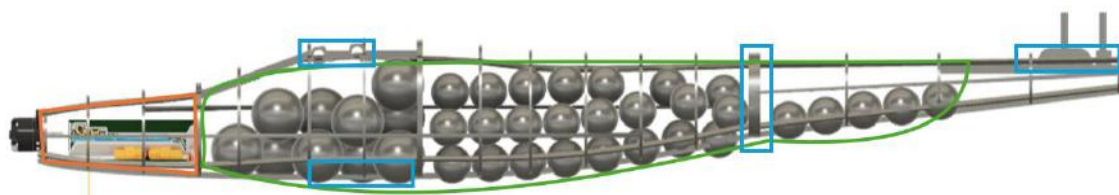


Figure 4: Preliminary fuselage structure. Avionics in orange, payload in green, connection points in blue.



Figure 5: Laser cutting ribs for wing assembly.



Figure 6: Wing assembly with balsa leading edge wrapped for drying.

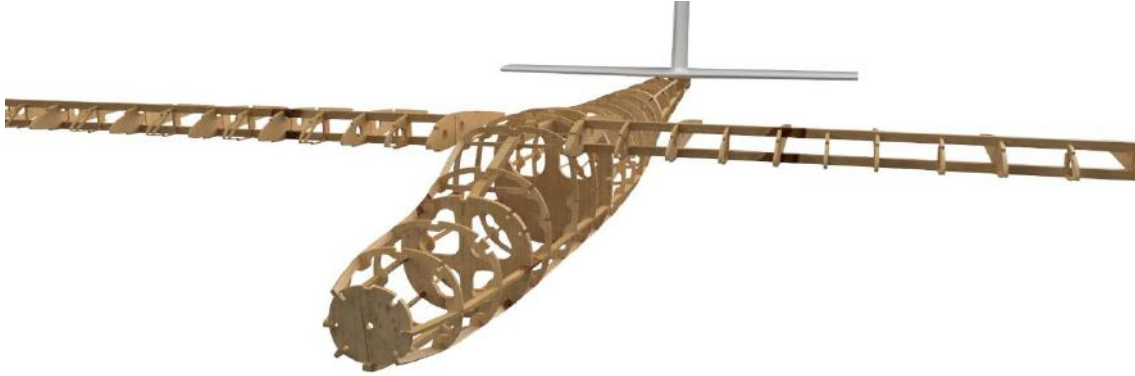


Figure 7: Partial assembly render of the ECLifter

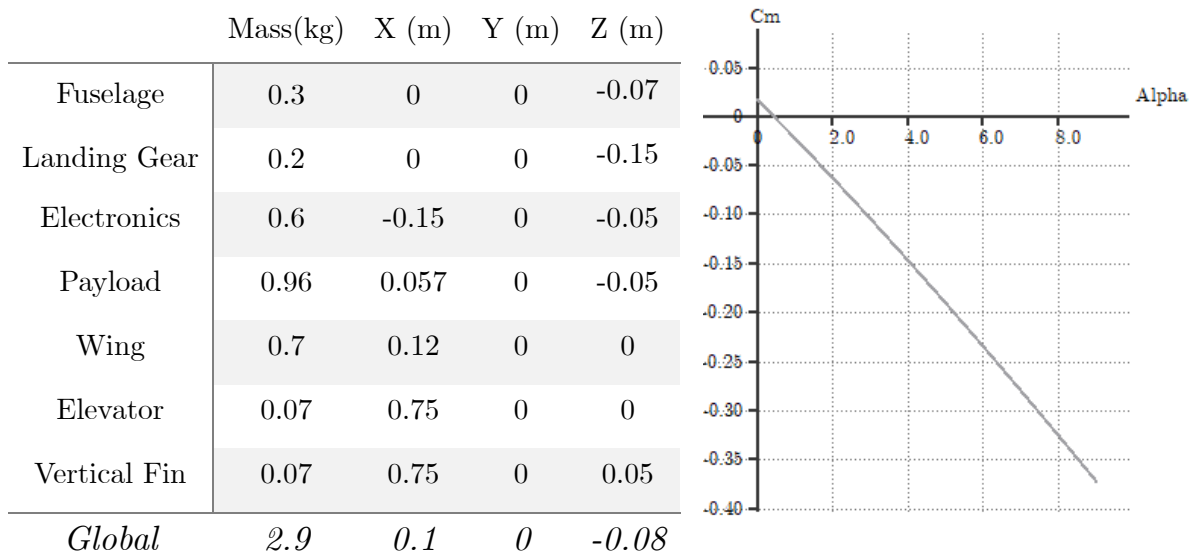


Figure 8: Mass estimation for a 6 ball PL and longitudinal static stability analysis. The center of the leading edge of the main wing is the origin of coordinates, with the positive direction being towards the back (X) and towards the top (Z) of the aircraft.